Spatial rescue in biofilms: a double-edged sword

May 11, 2026

Abstract

A biofilm under antibiotic treatment has a clear spatial asymmetry: cells on the surface are exposed to the drug, while cells in the deeper interior are shielded from it. We ask whether that shielded interior makes the population more likely to evolve resistance and survive (a process known as *evolutionary rescue*). By comparing a spatially structured biofilm model against a well-mixed population of the same initial size, and sweeping the antibiotic dose along with the metabolic activity of the interior, we find that the biofilm is a *double-edged sword*. When the interior is metabolically active, the biofilm rescues less often than the well-mixed reference at low dose, but more often at high dose. When the interior is dormant, the biofilm has no rescue advantage at any dose and is uniformly less likely to evolve resistance than the well-mixed reference population.

1 The question

The current antibiotic resistance crisis is a major public health threat, and biofilms are a key part of the problem. Biofilms are dense, surface-attached communities of bacteria that are common in nature and in infections. They are notoriously hard to kill with antibiotics, and they are thought to be responsible for a large fraction of chronic infections. Understanding how biofilms evolve resistance under antibiotic treatment is therefore an important question. Furthermore, treating infections with behavior-modifying interventions, such that a compound may induce increased biofilm formation, or increased dispersal (towards a well-mixed population), is an active area of research. The intuition is that while antibiotics induce high killing selective pressure, behavior-modifying interventions may not kill bacteria, but could reduce the population's ability to evolve antibiotic resistance by disrupting the spatial structure of the biofilm. To predict the evolutionary consequences of such interventions, we need to understand how the spatial structure of a biofilm affects the population's ability to evolve resistance in the first place, and whether applying such a treatment is likely to help or hurt the population's chances of survival when applied in conjunction with antibiotics.

Moreover, to treat bacterial infections effectively, we need to understand the evolutionary dynamics of resistance not in a simple well-mixed culture, but in the complex spatial structure of a biofilm. The spatial structure creates a natural asymmetry: cells on the surface are exposed to the drug, while cells in the deeper interior are shielded from it. This shielded interior is a spatial refuge that could potentially allow the population to survive longer under treatment, and therefore give it more time to evolve resistance. But does it actually help the population survive in the long run?

Thus, we set out to answer whether the structure of a biofilm actually helps the population survive in the long run, by allowing *evolutionary rescue*: the outcome in which a declining population avoids extinction because a resistant mutant arises and takes over in time.

The naive intuition is that the refuge provided by a biofilm should help a population evolve drug resistance. A shielded interior keeps the population around for longer timescales, so more mutations have a chance to arise. But more mutations is not the same thing as more rescue. For a mutation to matter, it has to reach the drug-exposed surface and grow there. Mutations stuck in the shielded interior do nothing on their own, because the interior carries no selection pressure to give a resistant cell its advantage. Whether the biofilm is a net help or a net hindrance to a population aiming to survive through evolving antibiotic resistance therefore depends on how effectively interior mutations make it to the surface, and on whether the spatial structure carries hidden costs that offset the supply benefit. That is what we set out to measure.

2 The model

The biofilm is a one-dimensional chain of cells. The outermost l cells of the chain form the **edge**, where the antibiotic is present. Everything deeper forms the **core**, where the antibiotic does not reach. Each cell is either wild-type (WT) or resistant (R). Resistance is irreversible: an R cell can only produce R daughters.

2.1 Birth and death rates

Each cell has a birth rate and a death rate that depend on its compartment and its genotype.

- **Edge, WT**: Birth rate $b_e = 1$, which sets the time unit. Death rate d_e is the antibiotic dose. We sweep over values $d_e > b_e$, so WT cells in the edge are always net-dying under treatment.
- **Edge, R**: Resistant cells in the edge have birth rate $b_R = 1$ and death rate $d_R = 0.6$. Their net growth rate $s_R = b_R - d_R = 0.4$ is positive, so once an R lineage establishes, it grows.
- **Core, WT and R**: Cells in the core have balanced birth and death rates, $b_c = d_c$, so the core has no net growth on its own. WT and R cells are treated identically in the core, since the shielded interior carries no selection pressure. The default core activity is $b_c = 0.2$, a slow but nonzero rate. A fully dormant interior corresponds to $b_c = 0$.

2.2 The 'conveyor belt' mechanism'

The core and edge are mechanically coupled by a simple bookkeeping rule: as long as the core still has cells in it, the edge always contains exactly l cells. Whenever an edge cell divides, the population grows by one and the innermost edge cell is reclassified as a core cell, to maintain the fixed edge length l. Whenever an edge cell dies, the population shrinks by one and the outermost core cell is reclassified as an edge cell. This rule produces two regimes of decline.

1. **Buffered decline** (phase of simulation where core is still present). When there there still cells left in the core (ie., $n(t) > l$), each death in the edge is immediately replaced by a cell drawn from the core, so the edge stays at l cells. Only the l edge cells contribute to net loss, and the total population falls linearly.

2. **Final collapse** (phase of simulation where core is gone). When the core is exhausted (ie., $N(t) \leq l$), the edge is the entire remaining population. There is no reservoir left to draw on, and the remaining cells decay exponentially, just as a well-mixed population would.

The buffered decline lasts a long time at low dose, when slow edge turnover gradually drains the core. It is brief at high dose, when fast edge turnover quickly burns through the core.

Mutations and rescue. Each birth event has a small probability μ of producing a resistant daughter instead of a wild-type one. Mutations can arise in either compartment. Because the core's birth rate is lower than the edge's, core cells generate mutations more slowly per unit time than edge cells do. A run counts as rescued once at least $r_{\mathsf{thr}} = 10$ resistant cells are present in the edge at the same time, a count large enough to virtually guarantee takeover.

The reference: a well-mixed population. To say whether the biofilm helps or hurts, we compare against a well-mixed population of the same initial size N_0. The well-mixed population uses only the edge rates: every cell is drug-exposed, with no shielded interior. This is the standard setting for evolutionary rescue.

3 Parameters and simulation conditions

Time is measured in edge generations ($b_e = 1$). We use a Gillespie stochastic simulation, with one event sampled at a time according to current rates.

Parameter	Value	Meaning
N_0	1000	Initial population size
l	100	Edge width (cells exposed to drug)
b_e	1.0	WT birth rate in edge (sets time unit)
d_e	dose (swept)	WT death rate in edge; super-MIC is $d_e > 1$
b_R	1.0	R birth rate in edge
d_R	0.6	R death rate in edge ($s_R = 0.4$)
$b_c = d_c$	0.2 (swept)	Core birth/death rate (active by default; 0 = dormant)
μ	10^{-4}	Mutation probability per birth
r_{thr}	10	Resistant edge cells required to declare rescue

Table 1: Default model parameters. The two swept variables are dose (d_e) and core activity (b_c).

We sweep dose over $d_e \in \{1.05, 1.1, 1.2, 1.4, 1.6, 1.9, 2.3, 2.8\}$ and core activity over $b_c \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8\}$, with $5{,}000$ independent runs at every parameter combination. The well-mixed reference is run at the same set of doses, also $5{,}000$ replicates per dose. Every run terminates naturally on either rescue ($r_{\mathsf{edge}} \geq r_{\mathsf{thr}}$) or extinction ($N = 0$); no artificial time or generation cap is applied.

4 The double-edged sword

To see whether the biofilm helps or hurts, we pick two doses at opposite ends of the swept range—a low dose just above the minimum inhibitory concentration ($d_e = 1.05$) and a high dose well above it ($d_e = 2.8$)—and compare the biofilm's rescue probability against the well-mixed population's at each. We run that comparison twice: once with a dormant core ($b_c = 0$), and once with a highly active core ($b_c = 0.8$). The four resulting comparisons are shown together in Figure 1.

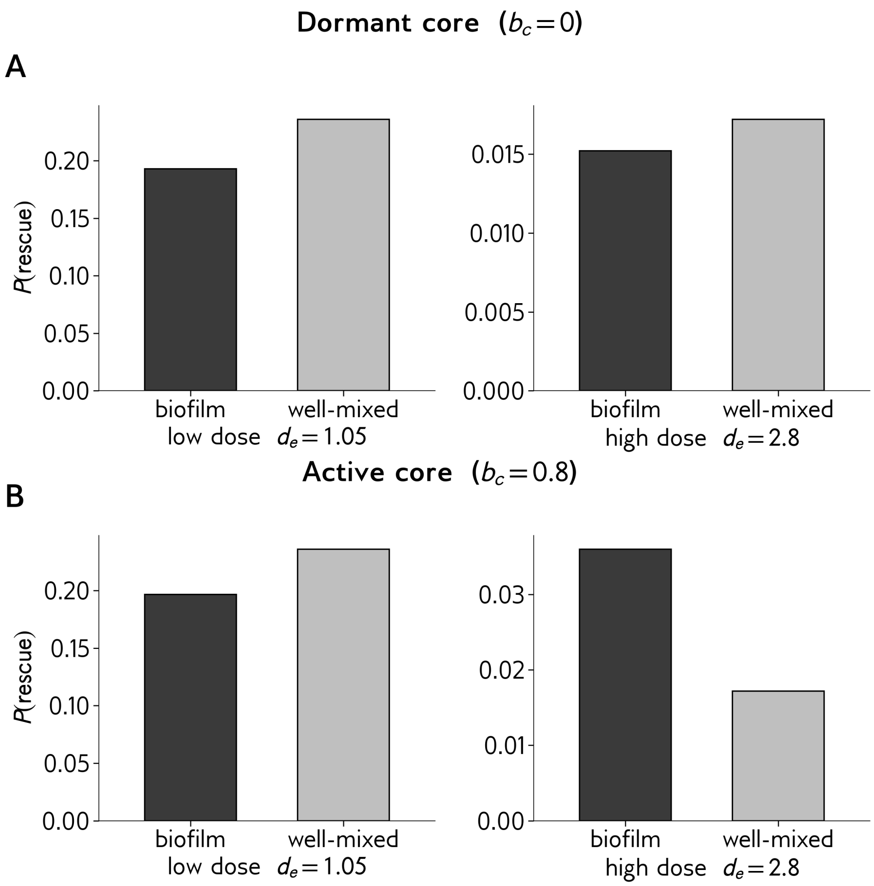

Figure 1: Rescue probability for the biofilm (dark) versus the well-mixed reference (light), at low dose ($d_e = 1.05$, left columns) and high dose ($d_e = 2.8$, right columns). **Panel A**: dormant core ($b_c = 0$). **Panel B**: active core ($b_c = 0.8$). Each subplot uses its own y-axis scale. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$; each bar averages over 5,000 independent simulation runs.

Panel A: dormant core. When the interior cannot divide, the biofilm rescues less often than the well-mixed population at both doses. The well-mixed bar is taller in each subplot. The shielded interior does not buy any benefit when nothing is happening inside it—the spatial structure is purely a liability.

Panel B: active core. When the interior is dividing at a steady rate, the picture splits in two. At low dose, the biofilm still loses—just as in Panel A. But at high dose the bars flip: the biofilm rescues roughly *twice* as often as the well-mixed reference. The same spatial structure that hurts at low dose now helps at high dose. This is the double-edged sword.

What this means together. The biofilm only wins under one specific combination: high dose plus an active core. Neither piece is enough on its own:

- A dormant core never helps, no matter the dose.
- An active core only helps at high dose; at low dose it still loses.

The two conditions have to coincide for the spatial structure to pay off.

This raises two follow-up questions. Why does the biofilm consistently lose at low dose, regardless of how active the core is? And what does an active core contribute at high dose that it does not contribute at low dose? The rest of the writeup works through these mechanisms.

5 Rescue probability across dose

The bar plot in Figure 1 sampled two specific doses at opposite ends of the swept range. To see whether the flip between them is a smooth crossover or just a jump between two unrelated regimes, we plot rescue probability across the full range of doses we swept (Figure 2).

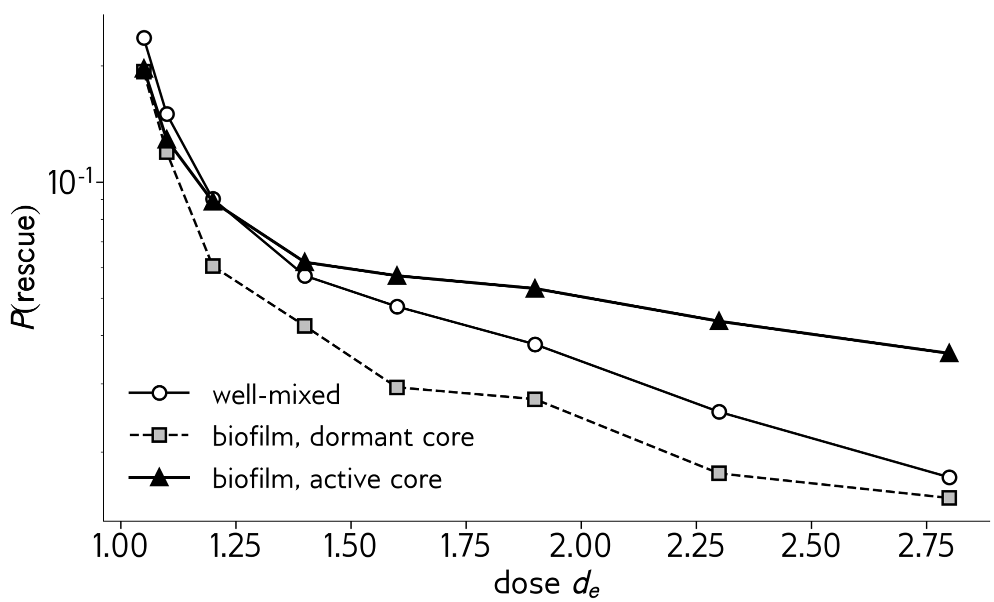

Figure 2: Rescue probability across dose, for the three conditions in Figure 1. The biofilm with a dormant core (dashed) stays below the well-mixed reference at every dose; no crossing exists. The biofilm with an active core (solid triangles) crosses above the well-mixed line around $d_e \approx 1.4$ and stays above it at higher doses. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$; each point averages over 5,000 runs.

The active-core biofilm crosses the well-mixed reference smoothly, at $d_e \approx 1.4$. Below that dose the biofilm loses; above it, the biofilm wins. The flip seen in the bar plot is one continuous crossover, not two separate effects.

The dormant-core biofilm does not cross at all. Its curve runs strictly below the well-mixed reference at every dose. The "double-edged sword" is therefore specifically a property of biofilms with metabolically active cores; a biofilm with a dormant interior is simply a worse rescue substrate than well-mixed at every dose.

Does the crossover depend on biofilm geometry? The curves above were all computed at one edge width, $l = 100$ cells out of $N_0 = 1000$. Real biofilms come in different thicknesses, and real antibiotics penetrate to different depths, so l is not a fixed quantity in the natural setting we are trying to model. Before trying to explain the crossover, it is worth checking whether it survives changes in l. Figure 3 sweeps l over a sixteenfold range while holding everything else fixed.

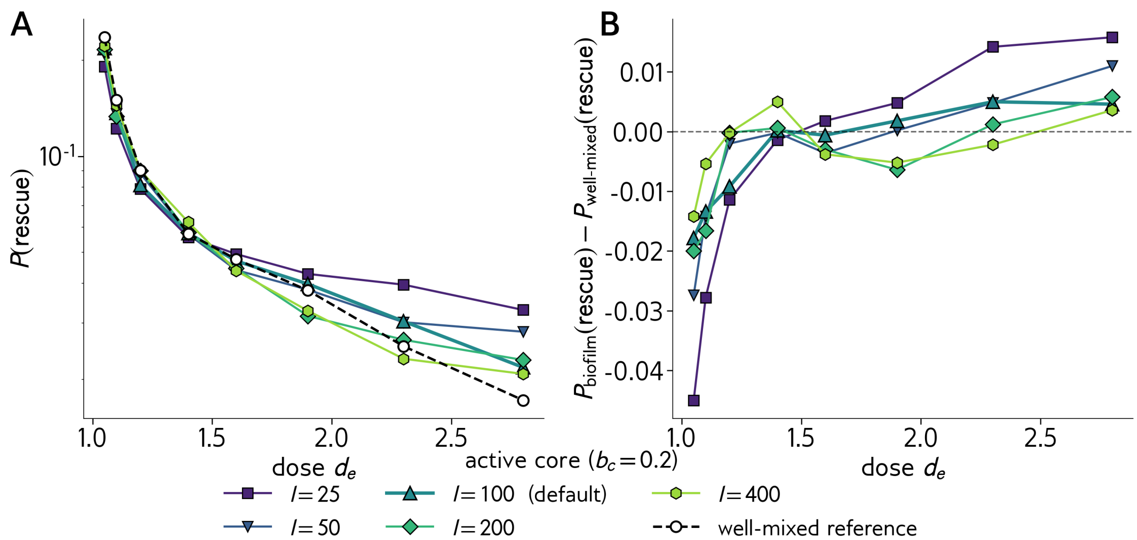

Figure 3: Two-panel view of the l sweep. **Panel A**: rescue probability across dose at every edge width in our sweep ($l \in \{25, 50, 100, 200, 400\}$) alongside the well-mixed reference. **Panel B**: the biofilm-minus-well-mixed difference $P_{\mathrm{BF}}(\text{rescue}) - P_{\mathrm{WM}}(\text{rescue})$. The crossover is the zero crossing in Panel B; the high-dose advantage is the height of each curve above zero. All five biofilm curves converge to the well-mixed value at low dose and diverge at high dose, with smaller l giving a larger high-dose biofilm advantage. The qualitative double-edge pattern is a structural feature of the model, not an artifact of one particular l. *Parameters*: $N_0 = 1000$, $b_c = 0.2$ (the sensitivity sweep's active-core value), $\mu = 10^{-4}$; each point averages over 5,000 runs.

The crossover survives. At every l in our sweep—a thin edge over a thick core ($l = 25$), a thick edge over a thin core ($l = 400$), and everything in between—the biofilm starts below the well-mixed reference at low dose and ends up above it at high dose. The crossover is structural, not a quirk of one specific geometry.

What does shift with l is the *location* of the crossover. Smaller l (a thinner edge sitting on top of a thicker core) pushes the crossover to a lower dose, and gives the biofilm a larger advantage at high dose. Larger l pushes the crossover later. Figure 4 summarizes this dependence by extracting one crossover dose per l.

The crossover dose increases with l. A thinner edge and a thicker core push the crossover earlier; a thicker edge and a thinner core delay it.

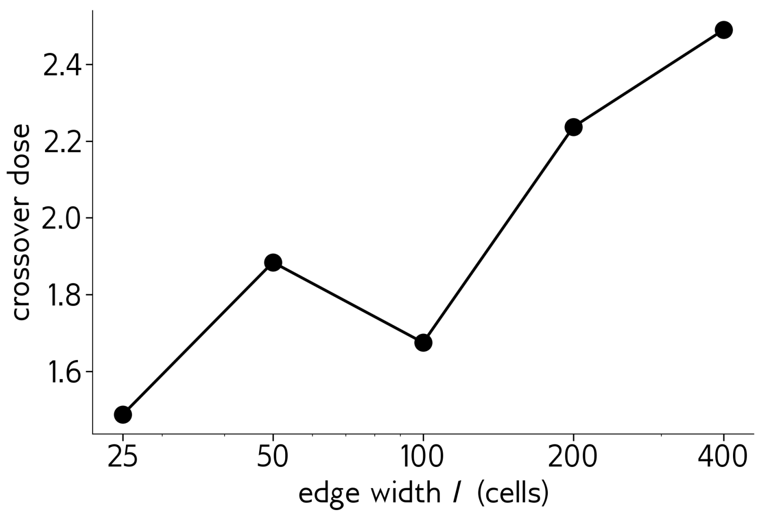

Figure 4: Crossover dose (the dose above which the biofilm rescues more often than the well-mixed reference at *every* higher dose) as a function of edge width l. For each l in Figure 3, we identify the largest sampled dose at which the biofilm is still at or below well-mixed, and linearly interpolate $P_{\mathrm{BF}} - P_{\mathrm{WM}}$ between that dose and the next-higher sampled dose (where the biofilm has gone above) to find the zero. The overall trend is increasing—thinner edges give earlier crossovers—and the small bump at $l = 50$ is consistent with sampling noise: the biofilm and well-mixed rescue probabilities differ by $\lesssim 0.004$ in the crossover region, near the resolution limit of 5,000 seeds. *Parameters*: same as Figure 3.

Does the crossover depend on how active the core is? Section 4 showed that a fully dormant core ($b_c = 0$) wipes out the biofilm's high-dose advantage entirely: the biofilm rescues less often than well-mixed at both ends of the dose range. An active core ($b_c = 0.8$) restored the advantage. We now sweep b_c across the intermediate range to find out how active the core actually has to be for the crossover to appear, and how the crossover dose moves with b_c. Figure 5 shows the dose curves and their well-mixed difference for the full b_c sweep.

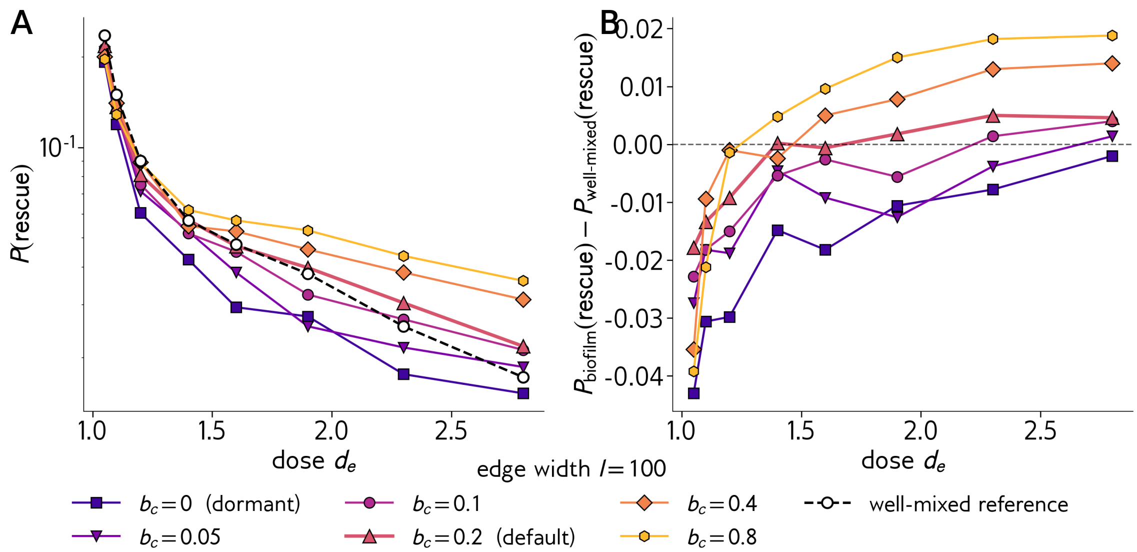

Figure 5: Two-panel view of the b_c sweep, at fixed $l = 100$. **Panel A**: rescue probability across dose at every core activity in our sweep ($b_c \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8\}$) alongside the well-mixed reference. **Panel B**: the biofilm-minus-well-mixed difference. The dormant ($b_c = 0$) curve sits strictly below zero at every dose—no crossover exists. Every other curve crosses zero somewhere in the swept range and sits above zero at high dose. The high-dose advantage grows monotonically with b_c. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$; each point averages over 5,000 runs.

The picture is clean: $b_c = 0$ has no crossover, while every positive b_c in the sweep does. Even a very slow core ($b_c = 0.05$, one-twentieth of the edge birth rate) is enough to produce a crossover, although it sits at a very high dose. As b_c increases, the crossover moves to lower doses and the high-dose advantage grows. Figure 6 summarizes the relationship by extracting one crossover dose per b_c, using the same strict definition as Figure 4.

The crossover dose drops sharply as the core wakes up, and then levels off. Going from $b_c = 0.05$ to $b_c = 0.2$ pulls the crossover from $d_e \approx 2.7$ down to $d_e \approx 1.7$; going from $b_c = 0.2$ to $b_c = 0.8$ only pulls it further down to $d_e \approx 1.25$. Once the core is active at all, the crossover is firmly inside the swept dose range; the precise speed of the core changes how easily the crossover is reached but does not gate the qualitative effect.

Why the crossover moves with l and b_c. The shifts in Figures 4 and 6 both go in the direction of *more core activity earns an earlier crossover*: a smaller l leaves more cells in the active interior, and a larger b_c makes each interior cell divide faster. Both knobs increase the same quantity—the rate at which the protected interior generates resistance mutations. Figure 7 makes the supply

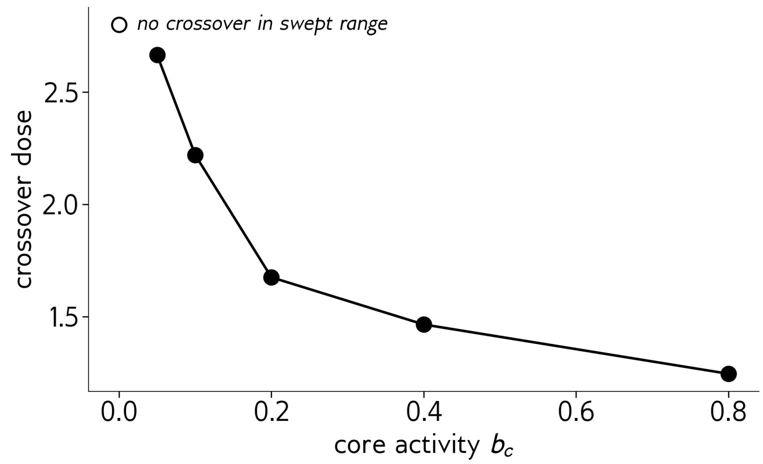

Figure 6: Crossover dose as a function of core activity b_c, with $l = 100$. The dormant case ($b_c = 0$) has no crossover in the swept range (open marker, top of plot). Every positive b_c has a finite crossover, and the crossover dose drops monotonically as the core gets more active—faster core turnover means the biofilm overtakes the well-mixed reference at a lower dose. *Parameters*: same as Figure 5.

effect explicit, and exposes a second, weaker effect that pushes the other way.

Both supply and delivery shift in opposite directions when we tune l or b_c. A smaller l produces a lot more mutations but delivers a smaller fraction of them; a larger b_c produces more mutations but delivers fewer. In both cases the supply effect wins by a large margin—the supply curves change by close to an order of magnitude across the swept range, while the delivery fractions change by a factor of two or three. The net effect is that crossovers happen earlier when either knob is turned toward "more active core", as we saw in Figures 4 and 6.

What is the biofilm doing differently from well-mixed? Rescue requires two things: a population has to *produce* a resistance mutation, and that mutation has to *take over* the edge. These two ingredients can be measured separately, so we can ask whether the biofilm differs from well-mixed in the rate at which mutations appear, the rate at which appearing mutations successfully establish, or both. Figure 8 plots the two ingredients side by side.

The two panels say opposing things.

Supply (Panel A): dormant biofilm and well-mixed lie on top of each other. The well-mixed curve and the biofilm-with-dormant-core curve are indistinguishable at every dose. Despite looking nothing alike dynamically—one is a steadily shrinking ball of cells, the other is a slowly draining biofilm with a protected interior—the two produce the *same number of resistance mutations* over their entire lifetimes. This is a real prediction of the model, not a coincidence, and Section 6 explains why. It also gives the dormant biofilm a useful role: it has the same mutation budget as well-mixed, so any difference in rescue rate between those two must come from how mutations *succeed*, not how many appear.

An active core changes the picture: the active-core curve sits about a factor of five above the other two. Those extra mutations come from cell divisions in the protected interior—divisions that

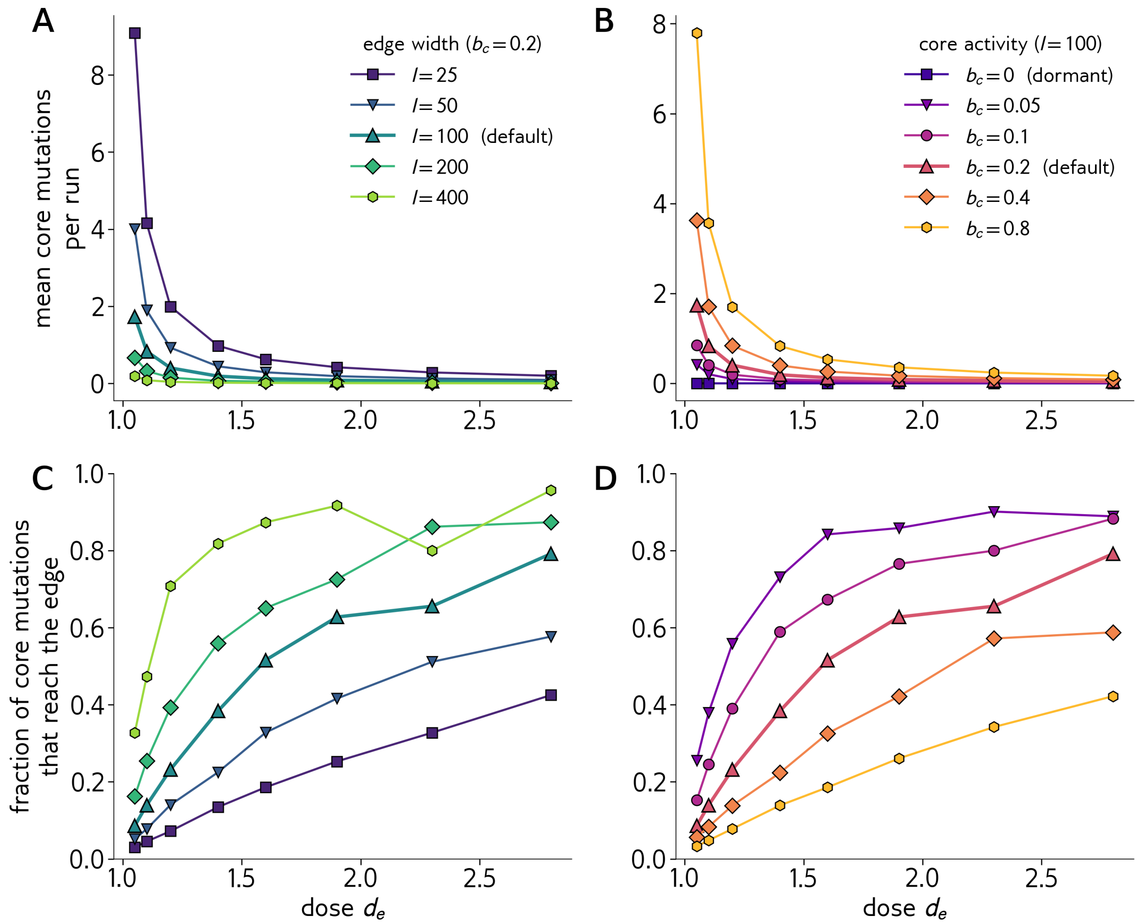

Figure 7: Two mechanism ingredients across the l sweep (left column) and the b_c sweep (right column). **Panels A, B**: mean number of core-born resistance mutations per run, vs dose. Smaller l and larger b_c both make this number much larger—the "supply bonus". **Panels C, D**: fraction of those core-born mutations that ever reach the edge (where they can grow). Smaller l *lowers* the delivery fraction at every dose: the boundary retreats at speed $s_e l$, so a thinner edge means a slower advance and more time for drift to kill core lineages. Larger b_c also lowers delivery: faster core turnover means more births competing for the same lineage's offspring, so each lineage faces more drift pressure. *Parameters*: $N_0 = 1000$, $\mu = 10^{-4}$; left column at $b_c = 0.2$, right column at $l = 100$; each point aggregates 5,000 runs.

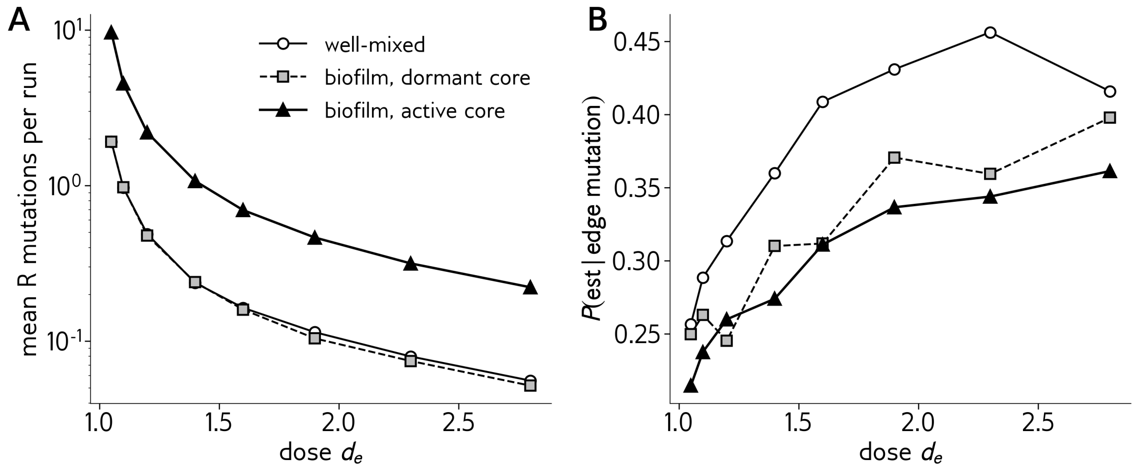

Figure 8: **A**: Mean number of resistance mutations produced per simulation run, across dose. The well-mixed and biofilm-dormant curves overlap almost exactly—an active core is what makes the biofilm generate extra mutations. **B**: Per-mutation establishment fraction, i.e. the fraction of edge-born mutations that grow to the establishment threshold. The well-mixed reference is highest at every dose; the biofilm-active is lowest at every dose. Each mutation in a biofilm has a worse chance of taking over. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$; each point averages over 5,000 runs.

the dormant biofilm and the well-mixed population do not have.

Establishment (Panel B): biofilms pay a per-mutation cost. The ordering reverses. The well-mixed curve is highest at every dose; the active-core biofilm is lowest; the dormant biofilm sits between them. Each individual resistance mutation in a biofilm is roughly $1.5\times$ less likely to take over than the same mutation in a well-mixed population. It is a real penalty, and it shows up regardless of whether the core is active.

The two panels together explain Figures 2 and 3. Rescue probability scales with the product of supply and establishment.

- The dormant biofilm has the same supply as well-mixed but pays the per-mutation cost, so it loses at every dose. No crossover is possible.
- The active biofilm pays the same per-mutation cost but receives a $\sim 5\times$ supply bonus. Whether the bonus outweighs the cost depends on dose: at low dose the cost wins, at high dose the bonus wins. The crossover is the dose at which they balance.

This already tells the qualitative story. What remains is to explain *where each effect comes from*: why an active core produces five times more mutations than well-mixed does, and why a biofilm mutation is harder to establish. The next section (Section 6) handles the supply story—it is the dominant effect and where the dose-dependent flip actually lives. The per-mutation cost is treated more briefly later (Section 8); it is a roughly constant tax in the background.

6 Where the mutations come from: lifetime and supply

The decomposition figure showed something surprising: well-mixed and the biofilm with a dormant core produce the same total mutation supply, even though the two populations look very different over time. The biofilm should live longer—it has a buffered interior. Why doesn't a longer lifetime translate into more mutations? And separately, where does an active core's roughly five-fold supply bonus actually come from? Both questions are best answered by looking at how the population moves through time. Figure 9 shows the two trajectories with mutations turned off, so the curves are pure wild-type decline.

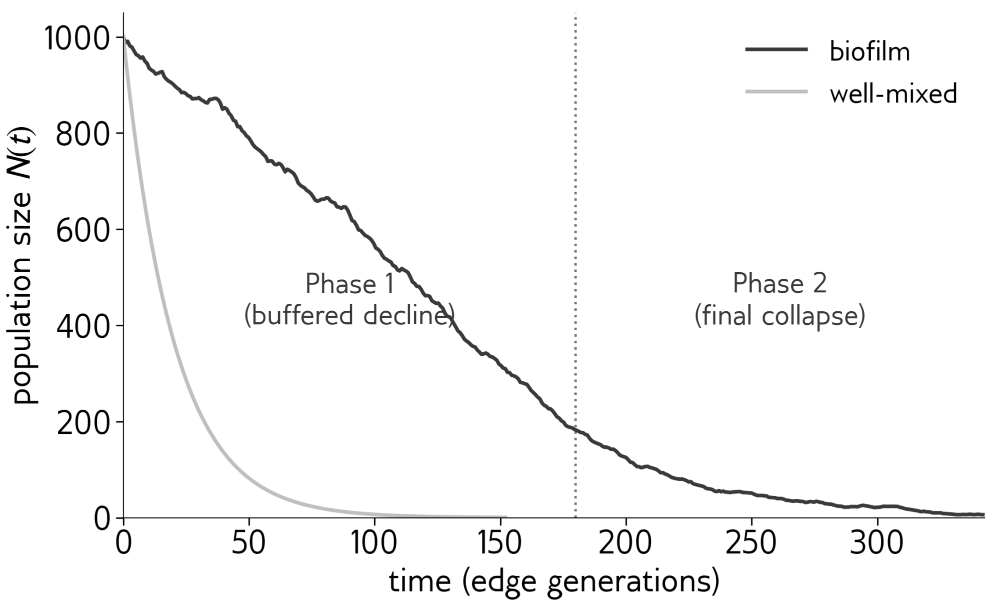

Figure 9: Population size $N(t)$ for the biofilm and the well-mixed reference, at $d_e = 1.05$ with mutations disabled. The biofilm declines linearly during Phase 1 (the core feeds replacement cells into the edge), then exponentially during Phase 2 (the core is exhausted). The well-mixed population declines exponentially throughout. The two curves have very different shapes but the same area underneath. *Parameters*: $N_0 = 1000$, $l = 100$, $b_c = 0.2$, $\mu = 0$ (no mutations, so what we see is the pure wild-type decline). Biofilm curve is the mean of 20 independent runs; well-mixed curve is the analytical exponential decay.

Two shapes, one area. The biofilm survives roughly twice as long as the well-mixed population, but it is at a smaller size for most of that extra time. Concretely: during Phase 1, only the $l = 100$ edge cells experience the antibiotic, so only those l cells contribute to the net decline. The total population falls linearly at rate $s_e l$ until the core is gone, which takes $T_1 = (N_0 - l)/(s_e l)$ time units. Once the core is exhausted, the remaining l cells decay exponentially at rate s_e for another $\sim \ln(l)/s_e$ time units before extinction.

What controls mutation supply is not lifetime; it is total cell-time, $\int N(t)\,dt$—the area under the curve. Each cell-time unit allows $b_e\mu$ wild-type birth-mutations on average, and the mutations are what the rescue race depends on. The well-mixed population is also dying exponentially at rate s_e, so its total cell-time is $\int_0^\infty N_0 e^{-s_e t}\, dt = N_0/s_e$. For the biofilm-with-dormant-core, only the edge contributes to mutation supply, and the edge holds l cells for time T_1 and then decays exponentially. Its total edge cell-time works out to

$$l \cdot \frac{N_0 - l}{s_e l} + \frac{l}{s_e} \;=\; \frac{N_0 - l}{s_e} + \frac{l}{s_e} \;=\; \frac{N_0}{s_e}.$$

Same integral, despite very different shapes. This is the clean prediction behind the curves overlapping in Figure 8A. A spatial structure that stretches a population's lifetime without adding any extra dividing cells does not change the mutation budget.

Where the active core's bonus comes from. An active core changes the picture by adding dividing cells—ones that are not contributing to the antibiotic-driven decline, but that are still hosting birth events. During Phase 1, core cells divide at rate b_c each, and there are roughly $(N_0 - l)/2$ of them on average over the Phase 1 window. Their integrated cell-time is

$$\text{core cell-time} \;\approx\; \frac{(N_0 - l)^2}{2 s_e l},$$

and the rate at which they produce mutations is $b_c \mu$ per cell. With our defaults at low dose ($N_0 = 1000$, $l = 100$, $s_e = 0.05$, $\mu = 10^{-4}$), the edge contributes roughly two mutations per run and the core, at $b_c = 0.8$, contributes another five or six. The biofilm with an active core therefore produces roughly three times the supply of the well-mixed reference, in line with what Figure 8A shows.

The active core's contribution is purely additive: it sits on top of the same edge supply that the well-mixed population also has. The biofilm does not gain or lose from the conveyor belt itself; it gains extra mutations exclusively because of the metabolic activity in the interior. A faster core means more core births, more mutations, and a bigger bonus.

So the supply story is settled. The biofilm-with-dormant-core and the well-mixed reference produce the same number of mutations because they share the same total edge cell-time, even though their lifetime profiles differ. A biofilm with an active core produces more mutations because its core cells are also dividing.

But supply alone does not determine rescue. The next section asks the follow-up question: of the extra mutations that an active core produces, how many are useful? It turns out most of them are not.

7 Why only some of those mutations matter

The previous section showed that an active core produces several extra resistance mutations per run. But producing a mutation is not the same as using it. A resistant cell only matters for rescue if it ends up in the edge—where it can grow under the antibiotic selection that gives it a survival advantage. A resistant cell stuck in the shielded core does nothing, because there is no selection there. So the question for this section is: of all the extra mutations an active core produces, how many actually make it out to the edge in time to matter?

Why mutations get stuck in the core. Resistant cells in the core grow and die at the same rate ($b_c = d_c$), so on average their numbers do not change. But "on average" hides what actually happens. A lineage that starts as a single cell mostly does not grow at all; it just bounces around randomly in size due to chance birth and death events, and eventually the bouncing brings it back to zero. This is sometimes called drift, and it is the default fate of a neutral lineage starting from one cell. To put numbers on it: a single neutral cell goes extinct with probability one given enough time.

For the lineage to escape this default fate, it has to get out of the core before bad luck eliminates it. The way out is mechanical: the biofilm is shrinking from the outside in, and the boundary between core and edge retreats inward at speed $s_e l$ per unit time. When the boundary passes over a core cell, that cell is reclassified as an edge cell. The lineage is now in the drug-exposed edge, where its resistance is worth something.

The wait between when a mutation is born and when the boundary catches up to it depends on the dose. At low dose the boundary retreats slowly, so the wait can be very long, giving the lineage a lot of time to die from drift before delivery. At high dose the boundary races inward, sweeping up surviving lineages before drift gets a chance.

Watching it happen (Figure 10). Figure 10 is a space-time picture of one such core-delivery rescue. Each row shows the state of the biofilm at one moment in time (time runs downward on the y-axis); each column shows one cell position, with the deep interior at the left and the outer surface at the right. Grey is wild-type core, tan is wild-type edge, blue is a resistant lineage. The black lines mark the inner edge of the core (left line) and the outer surface (right line). Both retreat leftward over time as the biofilm shrinks.

The figure shows the mechanism: the resistant cell appears in the core, sits there while bouncing around in size, gets reclassified into the edge once the boundary reaches it, and then takes over. Most of the action happens *not* when the mutation is created, but when it is delivered.

The aggregate signature (Figure 11A). Across many runs and many doses, the fraction of core-born resistant lineages that ever reach the edge climbs sharply with dose. Panel A of Figure 11 plots this fraction. For a highly active core ($b_c = 0.8$), the curve goes from about 5% at our lowest dose to about 40% at our highest dose—roughly an eightfold range. At the lower core activity ($b_c = 0.2$, the default in the main sweep), more lineages make it out at every dose, but the qualitative trend is the same. (The two values differ because faster core turnover means more births, so each lineage has more competing births nearby, but the underlying drift mechanism is identical.)

A small secondary effect: arrivals come as clumps, not single cells (Figure 11B). There is a second, weaker effect worth flagging. When a core lineage sits in the core for a long time before being delivered, it is not just drifting toward extinction—it can also drift toward larger sizes by the same coin flips. Conditional on *not* dying out, the surviving lineage tends to grow a bit. So lineages that wait a long time and survive arrive at the edge not as a single cell but as a small clump of cells. Panel B of Figure 11 shows this: arrival sizes average 1.5 cells at low dose (where the wait is long) and drop to 1.0–1.3 at high dose (where the wait is short). A clump of cells has a slight edge over a single cell because each cell is an independent chance to escape early-extinction drift, but the effect is modest (a factor of ~ 1.5 in clump size, not an order of magnitude), and

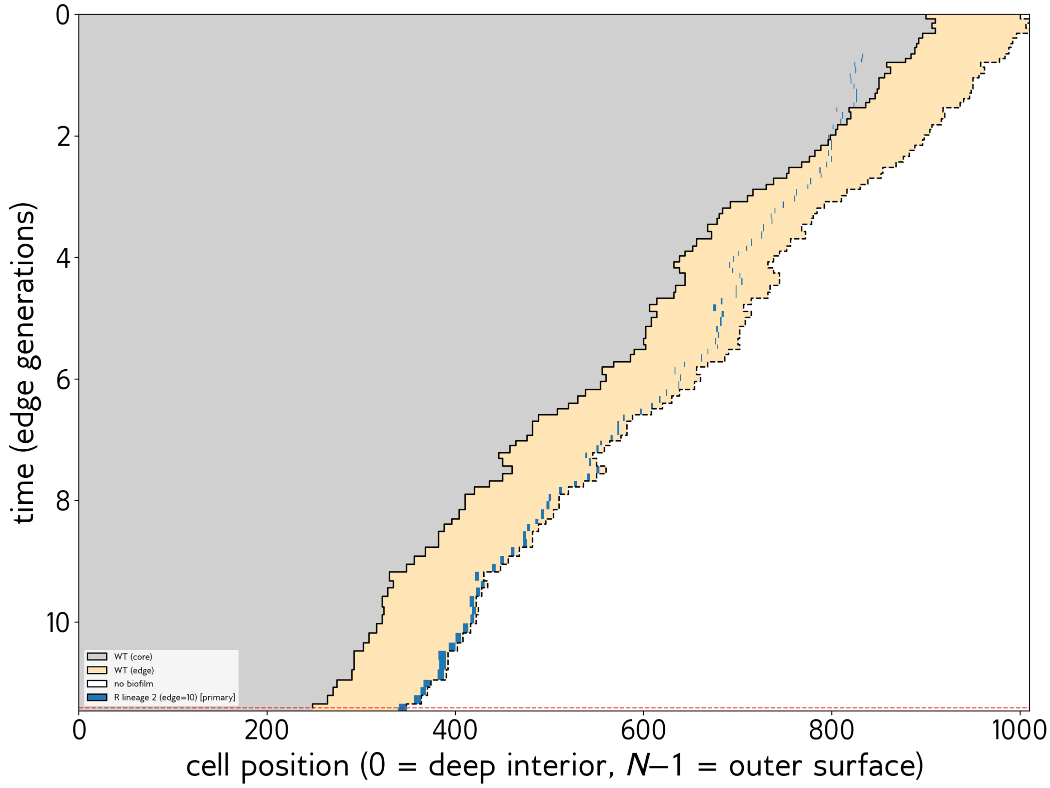

Figure 10: A core-delivery rescue. The resistant lineage (blue) appears in the shielded interior around position 800 at time $t \approx 0.7$. It does not grow much during its time in the core (where neither WT nor R has a selection advantage), but it survives. The boundary between core and edge—the upper-left black line—retreats inward toward the lineage. Around $t \approx 9$ the boundary reaches it and the resistant cells become edge cells. They then grow rapidly under antibiotic pressure, and rescue is triggered at $t = 11.4$. *Parameters*: single simulation run at $d_e = 1.5$, $N_0 = 1000$, $l = 100$, $b_c = 0.2$, $\mu = 5 \times 10^{-4}$. The mutation rate is higher than the default 10^{-4} used elsewhere, chosen so that a clean core-delivery rescue appears in a tractable seed scan; the mechanism does not depend on μ.

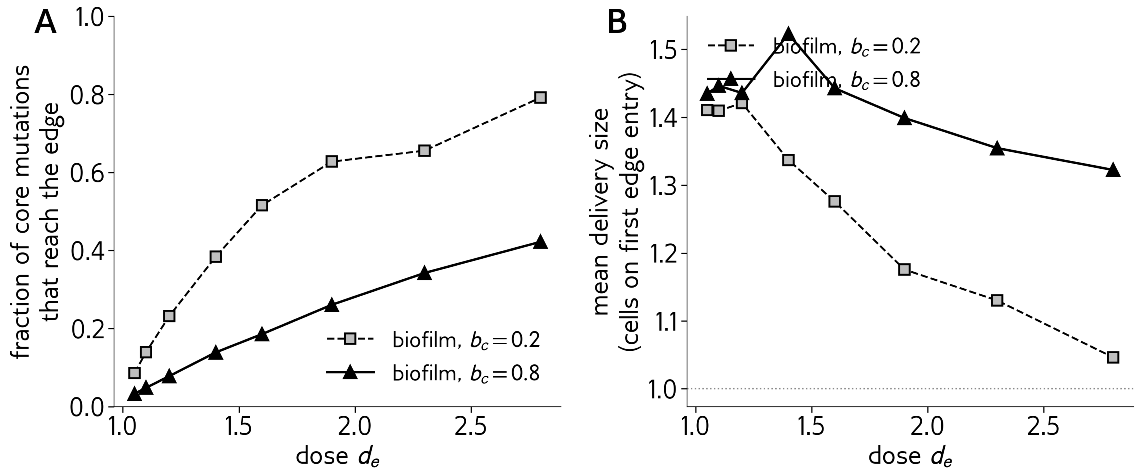

Figure 11: **A**: Fraction of core-born resistant lineages that ever reached the edge, plotted against dose, at two core activity levels. Both curves rise strongly with dose. At low dose almost all the core's extra mutations are lost before they can be useful; at high dose a much larger fraction reaches the edge. **B**: Average size of a delivered lineage at the moment it first enters the edge. A value of 1 (dotted line) means single-cell arrivals. Delivered lineages tend to arrive as small clumps of about 1–1.5 cells. Counterintuitively, the clumps are slightly *larger* at low dose, because the few lineages that survive the long wait have had more time to also grow. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$; each point aggregates all core-born mutations from 5,000 independent runs.

it points the opposite direction from the main delivery effect. The dominant story is the delivery fraction.

Putting Sections 6 and 7 together. An active core produces about five times the mutations of a well-mixed population at any dose. At low dose, almost none of those extra mutations get delivered into the edge in time, so they contribute little to rescue. The biofilm has effectively the same supply as well-mixed, and loses. At high dose, a substantial fraction of the active core's extra mutations is delivered before drift removes them, and the supply bonus finally translates into a rescue advantage. The crossover between these regimes is the bar-plot flip.

8 A small per-mutation tax

Brief treatment of the establishment penalty. Cite Panel B of Fig 8: BF mutations establish less often than WM. Cause is two-part: (i) global N is preserved by the core during Phase 1, lowering the density factor on R births; (ii) WT competitors in the edge are continually replaced. Either way, the result is a flat $\sim 20\%$ tax that the supply boost has to overcome.